Ashab Ali Khan

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Technical Skills

- **AI & Machine Learning:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Agentic AI, LangChain, LangGraph, Scikit-learn, TensorFlow, Keras.
- **Programming & Data:** Python, C++, SQL, NumPy, Pandas.
- **NLP & AI Tools:** NLP, Arabic NLP, Embeddings, Sentence Transformers, Semantic Search, Vector Search, ChromaDB, Groq API, Tavily, Tesseract OCR.
- **Tools & Deployment:** Docker, Streamlit, Hugging Face Spaces, Git, GitHub, Google Colab, REST APIs, API Integration.
- **Computer Science Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Software Engineering.

Projects

- **GCC AI & Startup Intelligence Platform (LIVE)** — Personal Project
  GitHub Repository — *Jun 2026*
  - Orchestrated end-to-end **multi-agent pipeline** using a **custom Python orchestration layer**, managing state passing between **6 sequential agents** from query intake to email delivery.
  - Architected a **6-agent sequential pipeline** processing over **30 web sources** across **6 GCC countries**, delivering professional Word reports via email in under **2 minutes**.
  - Constructed a **Query Understanding Agent** capable of classifying and routing **six specific query types**, enabling faster, more targeted research report generation.
  - **Assumed full ownership of scale optimization**; implemented batch processing of 6 sources per LLM call, resolving token limit crashes across the 30+ source data stream.
  - Engineered an **autonomous Review Agent** decision system, filtering **13-18 irrelevant entities** per query, thus enhancing the accuracy and relevance of generated reports.
  - **Expanded project scope from core logic to full-stack deployment**; streamlined the integration of Tavily, Groq LLM, SendGrid, and a Streamlit dashboard via Docker.
- **Arabic RAG Chatbot (LIVE)** — Personal Project
  GitHub Repository — *Jan 2026*
  - **Conceived and launched** a **bilingual Arabic-English RAG chatbot** for PDF querying using **LangChain, ChromaDB, and Arabic-Triplet-Matryoshka-V2 embeddings**.
  - **Diagnosed execution blockers**; and shifted to **Tesseract OCR** to bypass non-standard font encoding corruption across 81 pages.
  - **Eliminated** cross-lingual retrieval failure by **building an English-to-Arabic translation bridge**, aligning both languages into the same vector space.
  - Engineered **hybrid conversation memory** — last 2 full exchanges plus LLM-generated rolling summary — keeping token cost **constant at scale**.
  - **Deployed on Hugging Face Spaces** via Docker; resolved proxy-level 403 errors by disabling Streamlit XSRF protection in container launch configuration.
  - Implemented **per-message RTL rendering logic** storing language per conversation turn, preventing CSS alignment conflicts across mixed Arabic-English chat history.

Education

- **Galgotias University** — Greater Noida, UP
  Master of Computer Application (MCA) — *2023 - 2025*
  - **Core Focus Areas:** Machine Learning, Advanced Data Structures & Algorithms, Database Management Systems (DBMS).
- **Mahatma Jyotiba Phule Rohilkhand University (MJPRU)** — Bareilly, UP
  Bachelor of Science (BSc) — Physics, Chemistry, & Mathematics — *2019 - 2023*

Certifications

- **OpenAI Academy**
  - **Agents and Workflows** — Aug 2026
  - **Applied AI Foundations** — Aug 2026
  - **AI Foundations** — Aug 2026